



- 4 (a) By considering the binomial expansion of $\left(z + \frac{1}{z}\right)^6$, where $z = \cos \theta + i \sin \theta$, use de Moivre's theorem to show that

$$\cos^6 \theta = a \cos 6\theta + b \cos 4\theta + c \cos 2\theta + d,$$

where a, b, c and d are constants to be determined. [5]

[illegible]